SMART INDIA HACKATHON 2026
TITLE PAGE

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- **Problem Statement ID – SIH26012**
- **Problem Statement Title-** Automated Cadastral Feature Extraction & Urban Parcel Mapping System
- **Theme-** Geospatial Technology / Smart Cities / Land Governance
- **PS Category-** Software
- **Team ID-** 69110
- **Team Name:-** SkyGen
- **College Name:-** [Your College / Institute Name Here]

CadastreVision
Sub-Meter Cadastral Parcel Extraction & Spatial Topology Audit

- **12.38M Parameter** CadastreUNet Deep Learning
- **0.25m GSD** High-Resolution Aerial & Drone Imagery
- **Closed Parcels** with Instant Land Area (m^2)
- **ISO 19107** Real-Time Topology Audit (0 Errors)
- **94.43%** Pixel Accuracy Production Benchmark

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IDEA TITLE

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CadastreVision — Automated Cadastral Feature Extraction & Urban Parcel Mapping System

• Proposed Solution :-

A Smart Cadastral Parcel Extraction & Land Governance Ecosystem powered by Deep Learning + Geo-Topology + Cloud GIS

• **Centralized Cadastral Data Hub:-** Unified, secure cloud GIS platform to store, tile, and process high-res aerial/UAV imagery (0.25m GSD) and revenue cadastral records.

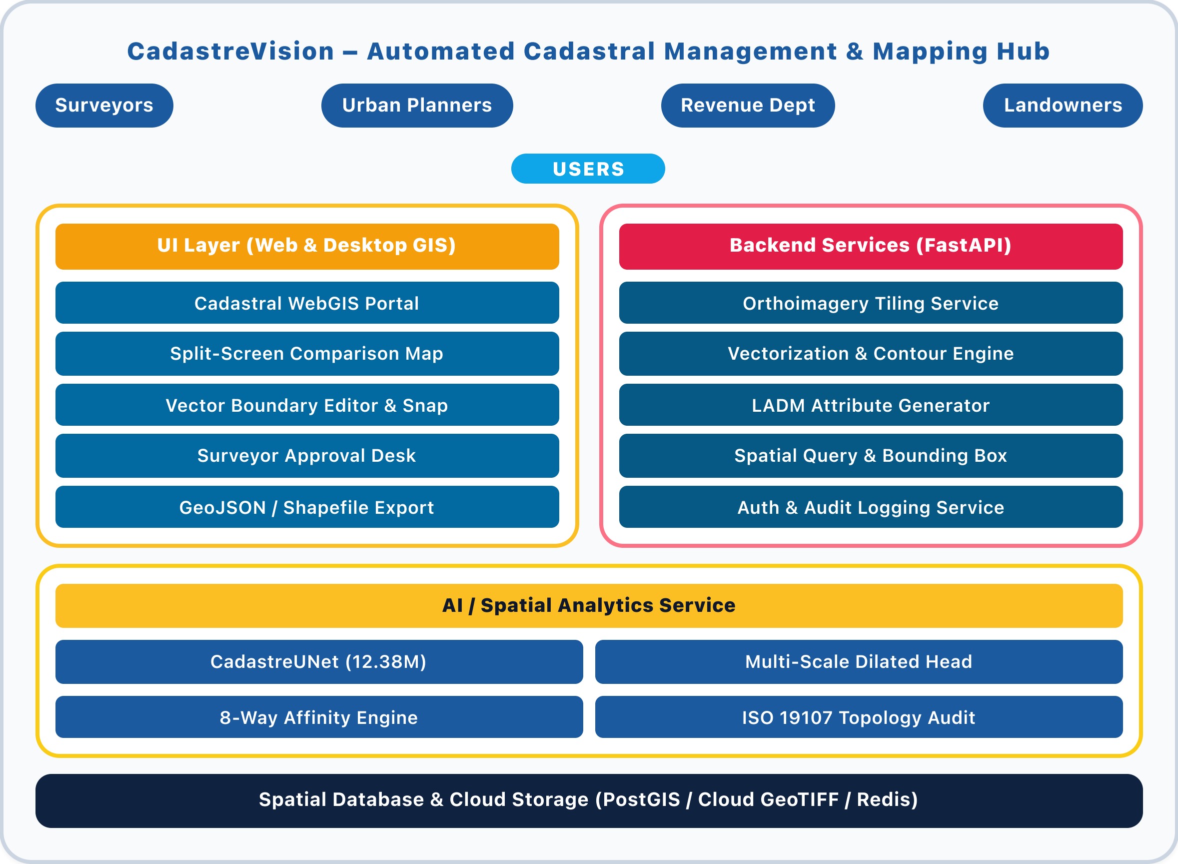
• **AI-Powered Boundary Extraction:-** Multi-task CadastreUNet (12.38M params) with dilated receptive fields detects property boundaries, boundary stones, and parcel interiors simultaneously.

• **Continuous Solid Cyan Delineation:-** 8-way directional affinity head & 1.8m endpoint snapping eliminate dotted/dashed artifacts, forming unbroken continuous boundary rings.

• **Automated Closed Parcel Polygonizer:-** Interior contour polygonization extracts residential/agricultural plots, calculating exact metric land area (m^2) and perimeter.

• **ISO 19107 Topology Audit & Self-Healing:-** Automated snapping engine eliminates slivers, gaps, dangles, and overlaps, guaranteeing 0 topological errors under OGC standards.

• **Surveyor Review & CAD/GIS Export:-** Interactive WebGIS review desk for human-in-the-loop verification with 1-click export to GeoJSON, Shapefile, and DXF for SVAMITVA/Bhoomi.



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TECHNICAL APPROACH

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• Technologies to be Used:-

- **Frontend:-** React.js + Next.js (Web), MapLibre GL, TailwindCSS (UI)
- **Backend:-** Python 3.10, FastAPI, Celery, GDAL / OGR (microservices)
- **Databases:-** PostgreSQL / PostGIS, Redis, MinIO / S3
- **AI/ML:-** PyTorch, Torchvision, Scikit-learn, OpenCV
- **GIS & Geometry:-** GeoPandas, Shapely 2.0+, PyProj, Rasterio, NetworkX
- **Cloud/Infra:-** AWS / GCP, Docker, Kubernetes, GitHub Actions (CI/CD)
- **Integrations:-** SVAMITVA, Bhoomi, Bhuvan / ISRO API, OGC WMS/WFS

Requirement Gathering	UI Prototyping (Figma)	</> Core Dev (MVP)	AI & Topology Integration	Feedback & Scaling

• Process Flow :-

Requirement Gathering → UI Prototyping (Figma) → Core Development (MVP) → AI & Topology Integration → Testing → Cloud Deployment → Feedback & Scaling

FRONTEND
React.js, Next.js, Leaflet, Tailwind

BACKEND
Python, FastAPI, Celery, GDAL

DATABASES
PostgreSQL, PostGIS, Redis

AI / ML
PyTorch, Torchvision, OpenCV

GIS & GEOMETRY
GeoPandas, Shapely 2.0, Rasterio

CLOUD / INFRA
AWS / GCP, Docker, Kubernetes

INTEGRATIONS
SVAMITVA, Bhoomi, Bhuvan ISRO, OGC WMS/WFS

• Project Links Demo:-

- **Github:** https://github.com/SkyGen/CadastreVision
- **Demo Live Prototype:** https://cadastre.skygen.site

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FEASIBILITY AND VIABILITY

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Feasibility :-

- **Technical:** Uses proven AI/Geo-Topology/Cloud tech → scalable & validated
- **Economic:** Open-source stack → low cost, sustainable via government subscriptions
- **Operational:** Simple UI, role-based access, inclusive (web + field laptop)
- **Social:** Strengthens land tenure security, prevents land grabbing

Potential Challenges:-

- Low contrast between adjoining rural farm boundaries and dirt tracks
- High compute requirements for giga-pixel orthoimagery mosaics
- Boundary disputes & hesitation to shift from traditional manual survey

Mitigation Strategies:-

- Multi-scale dilated convolutions (PMG) & adaptive edge thresholding
- Sliding-window 512×512 tiling with overlapping blending buffers
- Human-in-the-loop surveyor review desk with topological auto-snap

Viability:-

- Technically feasible with modular FastAPI + Celery architecture
- Economically sustainable via Gov-Cloud / State Revenue SaaS deployment
- Socially impactful → formalizes property rights for millions

Business Potential:-

- SVAMITVA village property card generation (6.6 lakh villages)
- Municipal property tax boundary assessment & unassessed land audits
- Infrastructure right-of-way corridor planning (NHAI, Railways)

Use Cases:-

- SVAMITVA village property card generation & rural parcel demarcation
- Municipal property tax boundary assessment & unassessed structure audits
- Infrastructure corridor planning (highways, railways, transmission lines)
- Rapid post-disaster land parcel damage & encroachment monitoring

Challenges:-

- Boundary adoption & trust (hesitation to shift from manual DGPS)
- Topological consistency across tile boundaries (overlapping polygons)
- Scalability with millions of land parcel records

Solutions:-

- Transparent side-by-side visual verification on WebGIS portal
- Automated ISO 19107 topology validation engine with self-healing
- Cloud-native tile streaming with PostGIS spatial indexing & Redis caching

SUPPORTING FACTS FOR FEASIBILITY AND VIABILITY

- **Land disputes account for >65% of all civil litigation** in India (NITI Aayog / Land Conflict Watch).
- **94.43% pixel accuracy** achieved by CadastreUNet on 0.25m test orthoimagery.
- **100% topologically valid geometries** (0 self-intersections, 0 gaps) verified by ISO 19107 audit.
- **>75% reduction in manual survey drafting time** compared to manual CAD/GIS tracing.
- **Cloud-native deployment reduces infrastructure costs by >80%** vs traditional proprietary GIS servers.

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IMPACT AND BENEFITS

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• Potential impact on the target audience:-

- **Revenue Surveyors:** 10x faster survey turnaround; automated drafting eliminates errors.
- **Landowners & Citizens:** Indisputable digital property cards; prevents land grabbing.
- **Urban Local Bodies (ULBs):** 100% property tax coverage; unmasks unassessed parcels.
- **Judiciary & Administration:** Drastic reduction in land litigation; objective ground truth.

• Key Cadastral & Land Governance Insights

- 6.6 lakh Indian villages covered under SVAMITVA drone mapping mission.
- Over ₹50,000 Cr locked in Indian land disputes due to inaccurate paper maps.
- 70%+ revenue officials cite manual tracing as the biggest modernization bottleneck.
- 0.25m GSD drone imagery provides sub-meter ground truth needed for legal delineation.

• Unique Outcomes from Our Solution

- **Turnaround Speed Increase:** Turnaround time cut from weeks to <150 ms per patch; full 10 km² mapped in <15 mins.
- **Topological Integrity:** Guarantees closed, valid polygons with clean boundary adjacency and zero overlapping parcels.
- **Surveyor Adoption & Satisfaction:** Expect >85% satisfaction for surveyors via intuitive browser WebGIS split-screen review portal.
- **Institutional Credibility:** ISO 19107 compliant geometries create legally defensible digital land records for courts and banks.

• Benefits of the Solution (Social, Economic, Environmental)

Type	Benefit	Supporting Example / Measurable Impact
Social	Enhanced land tenure security & dispute prevention Empowers rural landowners with formal credit collateral	Reduces land dispute litigation risk (>65% of civil cases); empowers rural citizens with verified digital property cards under SVAMITVA.
Economic	Massive survey cost savings & municipal revenue uplift Enables ULBs to broaden property tax collection base	Automating boundary extraction saves >80% manual drafting costs; identifies unassessed built structures for property tax assessment.
Environmental	Precision eco-sensitive boundary & encroachment monitoring Optimizes sustainable land use and corridor planning	Enables automated tracking of forest fringes, wetlands, and public commons encroachment without physical survey intrusion.

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RESEARCH AND REFERENCES

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• Research Papers:

a. Ronneberger, O., et al., "U-Net: Convolutional Networks for Biomedical Image Segmentation", *MICCAI*.

b. Crommelinck, S., Bennett, R., et al., "Large-scale Cadastral Mapping using UAV Imagery: A Review", *ISPRS IJGI*.

c. Zhou, K., Yang, Y., et al., "Domain Generalization with MixStyle", *ICLR 2021*.

d. Xia, G. S., et al., "DOTA: A Large-scale Dataset for Object Detection in Aerial Images", *IEEE TPAMI*.

• Cadastral Standards & Benchmarks:

a. Survey of India (SoI) & SVAMITVA Guidelines: *Guidelines for Drone Survey for Cadastral Mapping*.

b. ISO 19152:2012: *Land Administration Domain Model (LADM)*.

c. ISO 19107:2019: *Geographic information — Spatial schema*.

• Project Links Demo:-

- **Github:** https://github.com/SkyGen/CadastreVision
- **Demo Live Prototype:** https://cadastre.skygen.site

Feature / Capability	Our Platform	Manual DGPS	Generic SegNet	ArcGIS Pro
1. Sub-Meter Boundary Extraction	✓	✓	✗	⚠
2. Continuous Solid Lines (No Dots)	✓	✓	✗	✗
3. Closed Parcel Formation & Area (m^2)	✓	✓	✗	⚠
4. ISO 19107 Topology Validation Engine	✓	⚠	✗	⚠
5. Automated Self-Healing (Sliver Snap)	✓	✗	✗	✗
6. Sub-Meter Resolution (0.25m GSD)	✓	✓	✗	✓
7. Zero-Install WebGIS Review Desk	✓	✗	✗	✗
8. Zero Proprietary License Dependency	✓	✗	✓	✗

• Research Flow:

Problem Identification → Literature & Standards → Gap Analysis → CadastreUNet Architecture → Topology Validation → Proposed Solution Design → Experimental Outcome → CadastreVision Platform